

Final Report

Writing Template & Guideline

External Penetration Testing Assessment Report

Cover Page

Use the following cover page format for your final report:

External Penetration Testing Assessment Report

Prepared For:

PosBuzz Retail Solutions

Prepared By:

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Student ID:2026030000029

Serial Number:18

Course:

Applied Penetration Testing Engineer

Submission Date:18.5.26

```
Mon 18 May, 11:48 AttackBox IP:10.81.164.167

root@ip-10-81-164-167:~#

File Edit View Search Terminal Help

[*] 10.81.164.167:445 - Sending final
[*] 10.81.164.167:445 - Sending last
[*] 10.81.164.167:445 - Receiving res
[+] 10.81.164.167:445 - ETERNALBLUE C
(0xC000000D)!
[*] 10.81.164.167:445 - Sending egg t
[*] 10.81.164.167:445 - Triggering fr
[*] Sending stage (203846 bytes) to 10.81.164.167
[*] Meterpreter session 1 opened (10.81.164.167:445)
49236) at 2026-05-18 11:37:34 +0100
[+] 10.81.164.167:445 - ==-==-==-==-==-==
==-==-==-==-==-==
[+] 10.81.164.167:445 - ==-==-==-==-==-==
==-==-==-==-==-==
[+] 10.81.164.167:445 - ==-==-==-==-==-==
==-==-==-==-==-==

meterpreter > set RHOSTS 10.81.164.167
[-] Unknown command: set. Run the help command for more information.
meterpreter > set LHOST 10.81.146.38
[-] Unknown command: set. Run the help command for more information.
meterpreter > exploit
[-] Unknown command: exploit. Run the help command for more information.
meterpreter >
```

Table of Contents

Include a table of contents listing all sections with page numbers.

Section 1 — Executive Summary

Write a short summary of the entire assessment. Keep it concise — 3 to 5 sentences.

Example

A limited external penetration testing assessment was conducted against authorized TryHackMe training environments. Multiple security weaknesses were identified including exposed services, SMB vulnerabilities, and web application flaws. Remediation recommendations have been provided to improve the overall security posture.

Section 2 — Scope of Assessment

Clearly define what was included and excluded from this assessment.

In Scope

- Active Reconnaissance
- OWASP Juice Shop
- Blue Room

Out of Scope

- External systems
- Unauthorized targets

Section 3 — Methodology

Briefly explain the testing methodology used. Cover the following phases:

- Reconnaissance
- Enumeration
- Vulnerability Assessment
- Controlled Exploitation
- Documentation

Section 4 — Active Reconnaissance Summary

Document your findings from the Active Reconnaissance room. Include:

- Tasks completed
- Tools used
- Findings
- Security impact
- Remediation recommendations

Example topics to cover:

- Port scanning
- Service enumeration
- Exposed services

Section 5 — OWASP Juice Shop Summary

Document your web application findings from OWASP Juice Shop. Only 2–3 findings are required.

Include for each finding:

- Vulnerability identified
- Short explanation
- Security impact
- Remediation recommendation

Example findings:

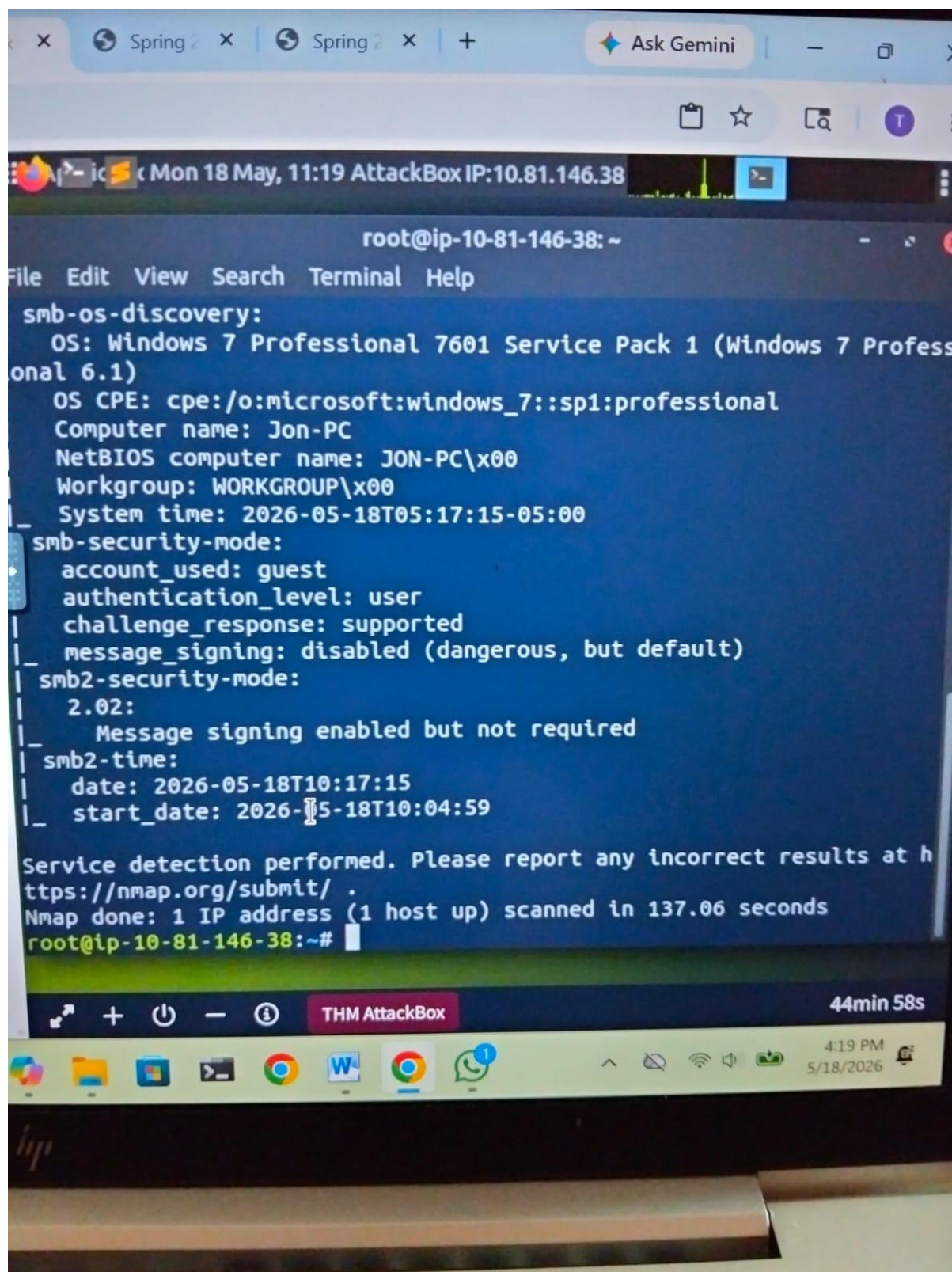
- SQL Injection
- Cross-Site Scripting (XSS)
- Weak Authentication

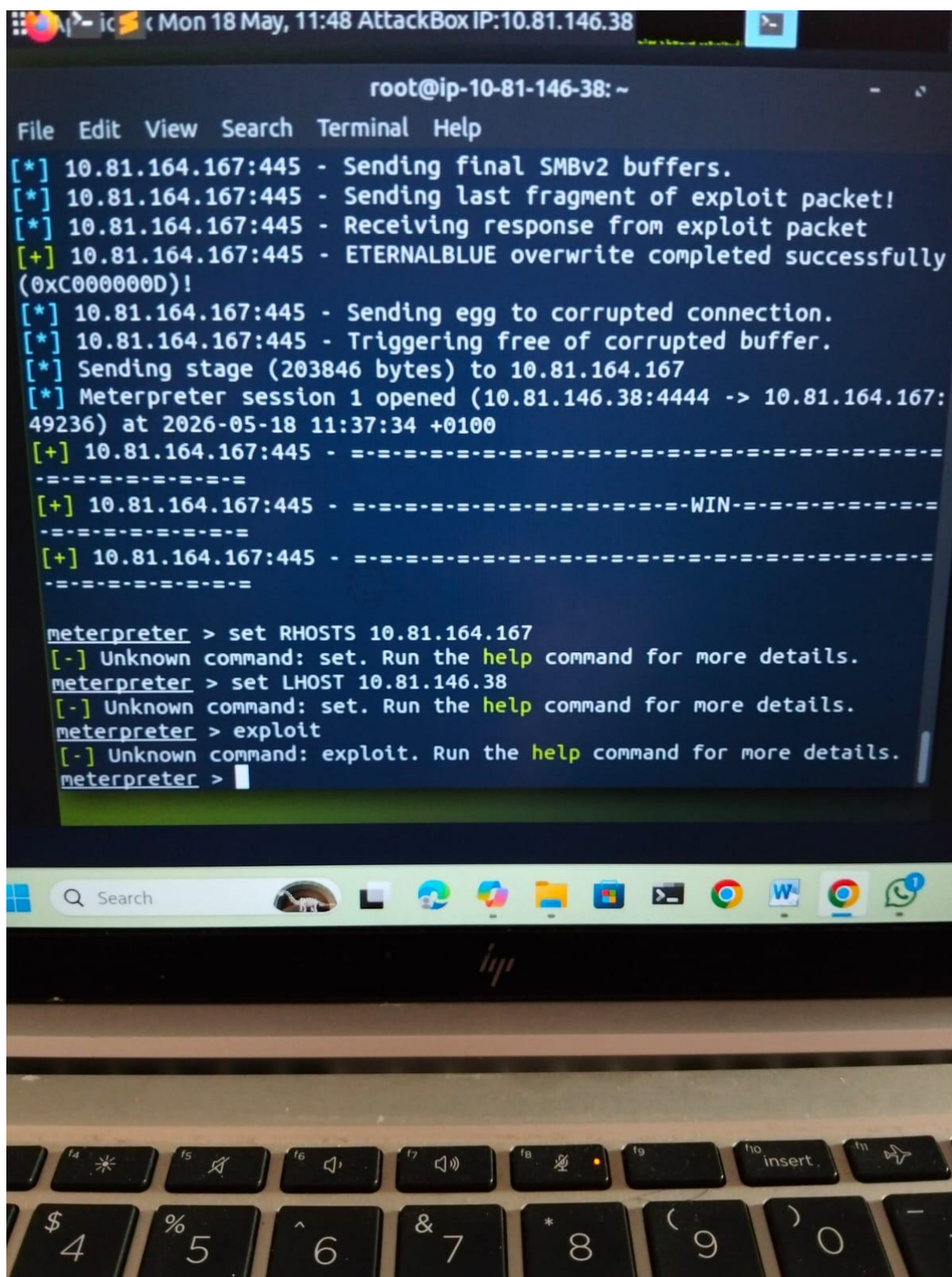
Section 6 — Blue Room Summary

Document your findings from the Blue Room exploitation exercise. Include:

- Vulnerability identified: The target system was found to be vulnerable to the EternalBlue [MS17-010] exploit, which allows for Remote Code Execution [RCE] via the SMB protocol.
- Exploitation summary: using the Metasploit Framework, the exploit/windows/smb/ms17_010_eternalblue module was configured with the target [RHOSTS] and attack box [LHOST] IP addresses. Successful exploitation resulted in a Metasploit session, providing administrative access to the system.
- Security impact: This vulnerability grants an attacker full control over the compromised machine, enabling unauthorized data access, system modification, and potential lateral movement within the network.

- Remediation recommendation:1.Apply the latest Microsoft security patches[MS17-010]too miltigate the vulnerability.
- 2.Disable the legacy and insecure SMBv1 protocol.
- 3.Restrict access to port 445 via firewall rules to allow traffic only from trusted sources.





Example topics:

- EternalBlue vulnerability
- SMB exposure
- Unauthorized access risk

Section 7 — General Remediation Recommendations

Provide overall remediation guidance. Examples:

- Apply security patches regularly
- Disable unnecessary services
- Restrict SMB access
- Implement strong password policies
- Use MFA where possible
- Validate user input properly
- Conduct regular vulnerability scans

Section 8 — Conclusion

Write a short closing paragraph summarizing the assessment outcome and the recommended next steps.

Example

The assessment identified several weaknesses within the assigned environments. Proper patching, service hardening, and secure coding practices would significantly improve the overall security posture.